

Supporting Information

Tabulated Values of the Shockley-Queisser Limit for Single Junction Solar Cells

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Table S1. Detailed balance limit for a single junction solar cell at a cell temperature T_c of 25 °C illuminated by the AM 1.5G spectral irradiance (ASTM G173-03) as a function of the energy band gap E_g and band gap wavelength λ_g considering radiative emission from the front and rear side of the solar cell as shown in Fig. 2a.

E_g [eV]	λ_g [nm]	$J_{\max}$ [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
0.31	3999.5	68.98	68.45	53.24	0.125	0.087	54.2	4.64
0.32	3874.5	68.69	68.31	53.93	0.133	0.094	55.6	5.06
0.33	3757.1	68.38	68.10	54.49	0.142	0.101	56.9	5.50
0.34	3646.6	68.06	67.87	54.98	0.150	0.108	58.2	5.93
0.35	3542.4	67.76	67.62	55.41	0.159	0.115	59.4	6.38
0.36	3444.0	67.45	67.35	55.75	0.167	0.122	60.5	6.82
0.37	3350.9	67.25	67.18	56.14	0.176	0.130	61.6	7.28
0.38	3262.7	67.15	67.10	56.56	0.185	0.137	62.6	7.76
0.39	3179.1	67.08	67.05	56.98	0.193	0.145	63.6	8.25
0.4	3099.6	66.95	66.93	57.29	0.202	0.152	64.5	8.73
0.41	3024.0	66.89	66.87	57.64	0.211	0.160	65.4	9.22
0.42	2952.0	66.81	66.80	57.94	0.220	0.168	66.3	9.72
0.43	2883.4	66.77	66.77	58.26	0.229	0.176	67.1	10.23
0.44	2817.8	66.77	66.77	58.56	0.237	0.184	67.8	10.75
0.45	2755.2	66.77	66.77	58.88	0.246	0.192	68.6	11.27
0.46	2695.3	66.77	66.77	59.15	0.255	0.200	69.2	11.80
0.47	2638.0	66.77	66.77	59.42	0.264	0.208	69.9	12.33
0.48	2583.0	66.77	66.77	59.66	0.273	0.216	70.5	12.86
0.49	2530.3	66.77	66.77	59.89	0.282	0.224	71.1	13.40
0.5	2479.7	66.75	66.74	60.08	0.291	0.232	71.7	13.94
0.51	2431.1	66.52	66.52	60.10	0.300	0.240	72.3	14.43
0.52	2384.3	66.20	66.20	59.99	0.309	0.248	72.8	14.89
0.53	2339.3	65.83	65.82	59.83	0.318	0.256	73.3	15.34
0.54	2296.0	65.36	65.36	59.57	0.327	0.265	73.8	15.76
0.55	2254.3	64.86	64.86	59.28	0.336	0.273	74.2	16.17
0.56	2214.0	64.32	64.32	58.94	0.345	0.281	74.7	16.56
0.57	2175.2	63.79	63.79	58.59	0.354	0.289	75.1	16.94
0.58	2137.7	63.24	63.24	58.22	0.363	0.298	75.5	17.32
0.59	2101.4	62.69	62.69	57.83	0.372	0.306	75.9	17.68
0.6	2066.4	62.20	62.20	57.50	0.381	0.314	76.3	18.06
0.61	2032.5	61.78	61.78	57.22	0.390	0.323	76.7	18.45
0.62	1999.7	61.53	61.53	57.09	0.399	0.331	77.1	18.89

E_g [eV]	λ_g [nm]	$J_{\max}$ [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
0.63	1968.0	61.17	61.17	56.86	0.408	0.339	77.4	19.30
0.64	1937.3	61.09	61.09	56.89	0.417	0.348	77.8	19.79
0.65	1907.4	61.08	61.08	56.98	0.426	0.357	78.1	20.32
0.66	1878.5	61.08	61.08	57.07	0.435	0.365	78.4	20.84
0.67	1850.5	61.08	61.08	57.16	0.445	0.374	78.7	21.37
0.68	1823.3	61.08	61.08	57.24	0.454	0.383	79.0	21.90
0.69	1796.9	61.03	61.03	57.27	0.463	0.392	79.3	22.41
0.7	1771.2	60.70	60.70	57.04	0.472	0.400	79.6	22.81
0.71	1746.3	60.17	60.17	56.61	0.481	0.409	79.9	23.13
0.72	1722.0	59.60	59.60	56.14	0.490	0.417	80.1	23.42
0.73	1698.4	58.97	58.97	55.61	0.499	0.426	80.4	23.67
0.74	1675.5	58.33	58.33	55.07	0.508	0.434	80.7	23.91
0.75	1653.1	57.68	57.68	54.52	0.518	0.443	80.9	24.14
0.76	1631.4	57.03	57.03	53.96	0.527	0.452	81.1	24.36
0.77	1610.2	56.38	56.38	53.40	0.536	0.460	81.4	24.56
0.78	1589.5	55.74	55.74	52.84	0.545	0.469	81.6	24.76
0.79	1569.4	55.12	55.12	52.30	0.554	0.477	81.8	24.96
0.8	1549.8	54.46	54.46	51.72	0.563	0.486	82.0	25.13
0.81	1530.7	53.81	53.81	51.16	0.572	0.495	82.2	25.30
0.82	1512.0	53.22	53.22	50.63	0.581	0.503	82.4	25.47
0.83	1493.8	52.73	52.73	50.21	0.590	0.512	82.6	25.70
0.84	1476.0	52.47	52.47	50.00	0.599	0.521	82.8	26.04
0.85	1458.6	52.29	52.29	49.87	0.609	0.530	83.0	26.41
0.86	1441.7	52.15	52.15	49.77	0.618	0.539	83.2	26.81
0.87	1425.1	52.09	52.09	49.75	0.628	0.548	83.4	27.24
0.88	1408.9	52.06	52.06	49.77	0.637	0.557	83.6	27.70
0.89	1393.1	52.06	52.06	49.80	0.646	0.566	83.7	28.17
0.9	1377.6	52.06	52.06	49.84	0.656	0.575	83.9	28.64
0.91	1362.5	52.06	52.06	49.87	0.665	0.584	84.1	29.11
0.92	1347.7	52.06	52.06	49.89	0.675	0.593	84.2	29.58
0.93	1333.2	51.83	51.83	49.71	0.684	0.602	84.4	29.91
0.94	1319.0	51.46	51.46	49.38	0.693	0.611	84.5	30.16
0.95	1305.1	50.98	50.98	48.95	0.703	0.620	84.7	30.33
0.96	1291.5	50.44	50.44	48.46	0.712	0.629	84.8	30.45
0.97	1278.2	49.87	49.87	47.94	0.721	0.637	85.0	30.55
0.98	1265.1	49.35	49.35	47.46	0.730	0.646	85.1	30.66
0.99	1252.4	48.80	48.80	46.96	0.739	0.655	85.3	30.75
1	1239.8	48.22	48.22	46.42	0.749	0.664	85.4	30.82
1.01	1227.6	47.65	47.65	45.90	0.758	0.673	85.5	30.88
1.02	1215.5	47.11	47.11	45.40	0.767	0.682	85.7	30.95
1.03	1203.7	46.61	46.61	44.94	0.776	0.691	85.8	31.03
1.04	1192.2	46.12	46.12	44.49	0.785	0.700	85.9	31.12
1.05	1180.8	45.66	45.66	44.06	0.795	0.709	86.0	31.21
1.06	1169.7	45.20	45.20	43.63	0.804	0.718	86.2	31.30

E_g [eV]	λ_g [nm]	J_{max} [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
1.07	1158.7	44.80	44.80	43.27	0.813	0.727	86.3	31.43
1.08	1148.0	44.56	44.56	43.06	0.823	0.736	86.4	31.66
1.09	1137.5	44.36	44.36	42.89	0.832	0.745	86.5	31.93
1.1	1127.1	44.23	44.23	42.77	0.842	0.754	86.6	32.23
1.11	1117.0	44.11	44.11	42.68	0.851	0.763	86.7	32.55
1.12	1107.0	43.81	43.81	42.40	0.860	0.772	86.9	32.73
1.13	1097.2	43.39	43.39	42.01	0.870	0.781	87.0	32.81
1.14	1087.6	42.92	42.92	41.57	0.879	0.790	87.1	32.84
1.15	1078.1	42.44	42.44	41.12	0.888	0.799	87.2	32.85
1.16	1068.8	41.95	41.95	40.66	0.898	0.808	87.3	32.85
1.17	1059.7	41.46	41.46	40.20	0.907	0.817	87.4	32.84
1.18	1050.7	40.97	40.97	39.73	0.916	0.826	87.5	32.81
1.19	1041.9	40.48	40.48	39.27	0.925	0.835	87.6	32.78
1.2	1033.2	39.99	39.99	38.80	0.935	0.844	87.7	32.74
1.21	1024.7	39.50	39.50	38.34	0.944	0.853	87.7	32.70
1.22	1016.3	39.01	39.01	37.89	0.953	0.862	87.8	32.65
1.23	1008.0	38.53	38.53	37.42	0.962	0.871	87.9	32.59
1.24	999.9	38.05	38.05	36.97	0.972	0.880	88.0	32.53
1.25	991.9	37.57	37.57	36.52	0.981	0.889	88.1	32.46
1.26	984.0	37.11	37.11	36.08	0.990	0.898	88.2	32.40
1.27	976.3	36.71	36.71	35.70	1.000	0.907	88.3	32.38
1.28	968.6	36.33	36.33	35.34	1.009	0.916	88.4	32.37
1.29	961.1	36.03	36.03	35.06	1.018	0.926	88.4	32.44
1.3	953.7	35.82	35.82	34.86	1.028	0.935	88.5	32.57
1.31	946.4	35.63	35.63	34.69	1.037	0.944	88.6	32.73
1.32	939.3	35.44	35.44	34.51	1.047	0.953	88.7	32.88
1.33	932.2	35.32	35.32	34.40	1.056	0.962	88.8	33.10
1.34	925.3	35.03	35.03	34.13	1.066	0.972	88.8	33.16
1.35	918.4	34.66	34.66	33.78	1.075	0.981	88.9	33.12
1.36	911.6	34.34	34.34	33.47	1.084	0.990	89.0	33.12
1.37	905.0	34.00	34.00	33.15	1.094	0.999	89.1	33.11
1.38	898.4	33.67	33.67	32.83	1.103	1.008	89.1	33.09
1.39	892.0	33.30	33.30	32.48	1.113	1.017	89.2	33.03
1.4	885.6	32.88	32.88	32.08	1.122	1.026	89.3	32.91
1.41	879.3	32.47	32.47	31.68	1.131	1.036	89.3	32.79
1.42	873.1	32.05	32.05	31.28	1.140	1.045	89.4	32.67
1.43	867.0	31.64	31.64	30.89	1.150	1.054	89.5	32.54
1.44	861.0	31.24	31.24	30.50	1.159	1.063	89.5	32.41
1.45	855.1	30.84	30.84	30.12	1.168	1.072	89.6	32.27
1.46	849.2	30.46	30.46	29.75	1.178	1.081	89.7	32.16
1.47	843.4	30.07	30.07	29.38	1.187	1.090	89.7	32.01
1.48	837.7	29.68	29.68	29.00	1.196	1.099	89.8	31.87
1.49	832.1	29.31	29.31	28.65	1.206	1.108	89.8	31.74
1.5	826.6	28.97	28.97	28.32	1.215	1.118	89.9	31.64
1.51	821.1	28.64	28.64	28.01	1.224	1.127	90.0	31.54

E_g [eV]	λ_g [nm]	J_{max} [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
1.52	815.7	28.33	28.33	27.71	1.234	1.136	90.0	31.46
1.53	810.4	27.99	27.99	27.38	1.243	1.145	90.1	31.34
1.54	805.1	27.63	27.63	27.02	1.253	1.154	90.1	31.18
1.55	799.9	27.26	27.26	26.67	1.262	1.163	90.2	31.02
1.56	794.8	26.90	26.90	26.33	1.271	1.172	90.3	30.86
1.57	789.7	26.55	26.55	25.98	1.281	1.182	90.3	30.69
1.58	784.7	26.19	26.19	25.63	1.290	1.191	90.4	30.51
1.59	779.8	25.83	25.83	25.28	1.299	1.200	90.4	30.33
1.6	774.9	25.47	25.47	24.94	1.309	1.209	90.5	30.14
1.61	770.1	25.12	25.12	24.60	1.318	1.218	90.5	29.95
1.62	765.3	24.82	24.82	24.31	1.327	1.227	90.6	29.82
1.63	760.6	24.67	24.67	24.17	1.337	1.237	90.6	29.88
1.64	756.0	24.40	24.40	23.90	1.346	1.246	90.7	29.77
1.65	751.4	24.05	24.05	23.57	1.356	1.255	90.7	29.57
1.66	746.9	23.71	23.71	23.24	1.365	1.264	90.8	29.37
1.67	742.4	23.38	23.38	22.92	1.374	1.273	90.8	29.17
1.68	738.0	23.06	23.06	22.61	1.384	1.283	90.9	28.98
1.69	733.6	22.74	22.74	22.30	1.393	1.292	90.9	28.80
1.7	729.3	22.46	22.46	22.02	1.402	1.301	91.0	28.64
1.71	725.1	22.19	22.19	21.76	1.412	1.310	91.0	28.50
1.72	720.8	21.91	21.91	21.50	1.421	1.319	91.1	28.35
1.73	716.7	21.67	21.67	21.26	1.430	1.329	91.1	28.23
1.74	712.6	21.37	21.37	20.96	1.440	1.338	91.2	28.03
1.75	708.5	21.06	21.06	20.67	1.449	1.347	91.2	27.82
1.76	704.5	20.76	20.76	20.37	1.458	1.356	91.3	27.62
1.77	700.5	20.47	20.47	20.09	1.468	1.365	91.3	27.42
1.78	696.5	20.18	20.18	19.81	1.477	1.374	91.3	27.22
1.79	692.6	19.91	19.91	19.54	1.487	1.384	91.4	27.03
1.8	688.8	19.65	19.65	19.29	1.496	1.393	91.4	26.86
1.81	685.0	19.40	19.40	19.05	1.505	1.402	91.5	26.70
1.82	681.2	19.11	19.11	18.77	1.515	1.411	91.5	26.48
1.83	677.5	18.83	18.83	18.49	1.524	1.420	91.5	26.26
1.84	673.8	18.54	18.54	18.22	1.533	1.430	91.6	26.03
1.85	670.2	18.27	18.27	17.95	1.543	1.439	91.6	25.81
1.86	666.6	17.99	17.99	17.68	1.552	1.448	91.7	25.59
1.87	663.0	17.72	17.72	17.41	1.561	1.457	91.7	25.37
1.88	659.5	17.46	17.46	17.16	1.571	1.466	91.7	25.15
1.89	656.0	17.22	17.22	16.92	1.580	1.475	91.8	24.96
1.9	652.5	16.97	16.97	16.68	1.589	1.485	91.8	24.75
1.91	649.1	16.72	16.72	16.43	1.599	1.494	91.9	24.54
1.92	645.8	16.47	16.47	16.19	1.608	1.503	91.9	24.33
1.93	642.4	16.22	16.22	15.95	1.617	1.512	91.9	24.11
1.94	639.1	15.97	15.97	15.71	1.627	1.522	92.0	23.89
1.95	635.8	15.73	15.73	15.47	1.636	1.531	92.0	23.67
1.96	632.6	15.49	15.49	15.24	1.645	1.540	92.0	23.45

E_g [eV]	λ_g [nm]	J_{max} [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
1.97	629.4	15.26	15.26	15.01	1.655	1.549	92.1	23.25
1.98	626.2	15.04	15.04	14.79	1.664	1.558	92.1	23.04
1.99	623.0	14.81	14.81	14.57	1.674	1.567	92.2	22.84
2	619.9	14.59	14.59	14.35	1.683	1.577	92.2	22.62
2.01	616.8	14.36	14.36	14.13	1.692	1.586	92.2	22.40
2.02	613.8	14.14	14.14	13.92	1.702	1.595	92.3	22.19
2.03	610.8	13.92	13.92	13.71	1.711	1.604	92.3	21.98
2.04	607.8	13.71	13.71	13.49	1.720	1.613	92.3	21.76
2.05	604.8	13.49	13.49	13.28	1.730	1.623	92.4	21.54
2.06	601.9	13.28	13.28	13.08	1.739	1.632	92.4	21.33
2.07	599.0	13.08	13.08	12.87	1.748	1.641	92.4	21.12
2.08	596.1	12.87	12.87	12.68	1.758	1.650	92.5	20.91
2.09	593.2	12.67	12.67	12.48	1.767	1.660	92.5	20.71
2.1	590.4	12.48	12.48	12.29	1.776	1.669	92.5	20.50
2.11	587.6	12.29	12.29	12.11	1.786	1.678	92.5	20.31
2.12	584.8	12.10	12.10	11.91	1.795	1.687	92.6	20.10
2.13	582.1	11.90	11.90	11.72	1.804	1.696	92.6	19.87
2.14	579.4	11.71	11.71	11.53	1.814	1.706	92.6	19.66
2.15	576.7	11.52	11.52	11.35	1.823	1.715	92.7	19.46
2.16	574.0	11.34	11.34	11.17	1.832	1.724	92.7	19.25
2.17	571.4	11.15	11.15	10.99	1.842	1.733	92.7	19.04
2.18	568.7	10.97	10.97	10.81	1.851	1.742	92.8	18.84
2.19	566.1	10.80	10.80	10.64	1.861	1.752	92.8	18.63
2.2	563.6	10.62	10.62	10.47	1.870	1.761	92.8	18.42
2.21	561.0	10.44	10.44	10.29	1.879	1.770	92.8	18.22
2.22	558.5	10.27	10.27	10.13	1.889	1.779	92.9	18.01
2.23	556.0	10.10	10.10	9.96	1.898	1.789	92.9	17.81
2.24	553.5	9.93	9.93	9.79	1.907	1.798	92.9	17.60
2.25	551.0	9.76	9.76	9.62	1.917	1.807	93.0	17.39
2.26	548.6	9.60	9.60	9.46	1.926	1.816	93.0	17.18
2.27	546.2	9.43	9.43	9.30	1.935	1.825	93.0	16.97
2.28	543.8	9.27	9.27	9.14	1.945	1.835	93.0	16.76
2.29	541.4	9.11	9.11	8.98	1.954	1.844	93.1	16.56
2.3	539.1	8.96	8.96	8.84	1.963	1.853	93.1	16.37
2.31	536.7	8.80	8.80	8.68	1.973	1.862	93.1	16.16
2.32	534.4	8.64	8.64	8.53	1.982	1.871	93.1	15.95
2.33	532.1	8.50	8.50	8.38	1.991	1.881	93.2	15.76
2.34	529.8	8.34	8.34	8.23	2.001	1.890	93.2	15.55
2.35	527.6	8.19	8.19	8.08	2.010	1.899	93.2	15.34
2.36	525.4	8.05	8.05	7.95	2.019	1.908	93.2	15.16
2.37	523.1	7.91	7.91	7.80	2.029	1.917	93.3	14.95
2.38	520.9	7.76	7.76	7.66	2.038	1.927	93.3	14.76
2.39	518.8	7.63	7.63	7.53	2.047	1.936	93.3	14.57
2.4	516.6	7.50	7.50	7.41	2.057	1.945	93.3	14.40
2.41	514.5	7.37	7.37	7.27	2.066	1.954	93.4	14.21

E_g [eV]	λ_g [nm]	J_{max} [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
2.42	512.3	7.24	7.24	7.14	2.075	1.963	93.4	14.02
2.43	510.2	7.10	7.10	7.01	2.084	1.973	93.4	13.82
2.44	508.1	6.96	6.96	6.88	2.094	1.982	93.4	13.62
2.45	506.1	6.83	6.83	6.74	2.103	1.991	93.5	13.42
2.46	504.0	6.70	6.70	6.62	2.112	2.000	93.5	13.23
2.47	502.0	6.58	6.58	6.49	2.122	2.009	93.5	13.04
2.48	499.9	6.45	6.45	6.37	2.131	2.018	93.5	12.86
2.49	497.9	6.33	6.33	6.25	2.140	2.028	93.6	12.66
2.5	495.9	6.20	6.20	6.12	2.150	2.037	93.6	12.47
2.51	494.0	6.07	6.07	6.00	2.159	2.046	93.6	12.27
2.52	492.0	5.95	5.95	5.88	2.168	2.055	93.6	12.08
2.53	490.1	5.83	5.83	5.76	2.177	2.064	93.6	11.89
2.54	488.1	5.72	5.72	5.65	2.187	2.073	93.7	11.71
2.55	486.2	5.61	5.61	5.54	2.196	2.083	93.7	11.54
2.56	484.3	5.50	5.50	5.43	2.205	2.092	93.7	11.36
2.57	482.4	5.38	5.38	5.32	2.214	2.101	93.7	11.17
2.58	480.6	5.27	5.27	5.20	2.224	2.110	93.8	10.98
2.59	478.7	5.15	5.15	5.09	2.233	2.119	93.8	10.78
2.6	476.9	5.04	5.04	4.98	2.242	2.128	93.8	10.59
2.61	475.0	4.93	4.93	4.87	2.251	2.138	93.8	10.40
2.62	473.2	4.82	4.82	4.76	2.261	2.147	93.8	10.22
2.63	471.4	4.71	4.71	4.65	2.270	2.156	93.9	10.03
2.64	469.6	4.61	4.61	4.55	2.279	2.165	93.9	9.85
2.65	467.9	4.50	4.50	4.45	2.288	2.174	93.9	9.67
2.66	466.1	4.40	4.40	4.35	2.298	2.183	93.9	9.49
2.67	464.4	4.30	4.30	4.25	2.307	2.192	93.9	9.31
2.68	462.6	4.20	4.20	4.15	2.316	2.201	94.0	9.13
2.69	460.9	4.09	4.09	4.05	2.325	2.210	94.0	8.94
2.7	459.2	4.00	4.00	3.95	2.334	2.219	94.0	8.77
2.71	457.5	3.90	3.90	3.85	2.343	2.229	94.0	8.59
2.72	455.8	3.80	3.80	3.76	2.353	2.238	94.0	8.41
2.73	454.2	3.71	3.71	3.67	2.362	2.247	94.0	8.23
2.74	452.5	3.62	3.62	3.58	2.371	2.256	94.1	8.07
2.75	450.9	3.52	3.52	3.48	2.380	2.265	94.1	7.89
2.76	449.2	3.43	3.43	3.39	2.389	2.274	94.1	7.71
2.77	447.6	3.34	3.34	3.31	2.398	2.283	94.1	7.54
2.78	446.0	3.26	3.26	3.22	2.408	2.292	94.1	7.39
2.79	444.4	3.18	3.18	3.14	2.417	2.301	94.2	7.23
2.8	442.8	3.10	3.10	3.06	2.426	2.310	94.2	7.08
2.81	441.2	3.02	3.02	2.99	2.435	2.319	94.2	6.92
2.82	439.7	2.95	2.95	2.91	2.444	2.328	94.2	6.78
2.83	438.1	2.88	2.88	2.85	2.453	2.337	94.2	6.65
2.84	436.6	2.81	2.81	2.78	2.463	2.346	94.2	6.51
2.85	435.0	2.73	2.73	2.70	2.472	2.355	94.3	6.37
2.86	433.5	2.67	2.67	2.64	2.481	2.364	94.3	6.24

E_g [eV]	λ_g [nm]	J_{max} [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
2.87	432.0	2.60	2.60	2.58	2.490	2.374	94.3	6.11
2.88	430.5	2.55	2.55	2.53	2.499	2.383	94.3	6.02
2.89	429.0	2.51	2.51	2.48	2.509	2.392	94.3	5.93
2.9	427.5	2.45	2.45	2.42	2.518	2.401	94.3	5.81
2.91	426.1	2.39	2.39	2.36	2.527	2.410	94.4	5.69
2.92	424.6	2.33	2.33	2.30	2.536	2.419	94.4	5.56
2.93	423.2	2.26	2.26	2.24	2.545	2.428	94.4	5.44
2.94	421.7	2.20	2.20	2.18	2.555	2.437	94.4	5.31
2.95	420.3	2.14	2.14	2.12	2.564	2.446	94.4	5.19
2.96	418.9	2.09	2.09	2.07	2.573	2.455	94.4	5.07
2.97	417.5	2.03	2.03	2.01	2.582	2.464	94.5	4.95
2.98	416.1	1.97	1.97	1.95	2.591	2.473	94.5	4.82
2.99	414.7	1.91	1.91	1.89	2.600	2.482	94.5	4.70
3	413.3	1.86	1.86	1.84	2.609	2.491	94.5	4.58
3.01	411.9	1.80	1.80	1.78	2.618	2.500	94.5	4.46
3.02	410.5	1.75	1.75	1.73	2.627	2.509	94.5	4.35
3.03	409.2	1.70	1.70	1.68	2.636	2.518	94.6	4.24
3.04	407.8	1.65	1.65	1.63	2.645	2.527	94.6	4.12
3.05	406.5	1.60	1.60	1.58	2.654	2.536	94.6	4.01
3.06	405.2	1.55	1.55	1.53	2.664	2.545	94.6	3.90
3.07	403.9	1.50	1.50	1.48	2.673	2.554	94.6	3.79
3.08	402.5	1.45	1.45	1.43	2.682	2.563	94.6	3.68
3.09	401.2	1.40	1.40	1.39	2.690	2.572	94.6	3.56
3.1	399.9	1.35	1.35	1.34	2.699	2.581	94.7	3.45
3.11	398.7	1.31	1.31	1.29	2.708	2.590	94.7	3.35
3.12	397.4	1.27	1.27	1.26	2.718	2.599	94.7	3.27
3.13	396.1	1.25	1.25	1.24	2.727	2.608	94.7	3.23
3.14	394.9	1.22	1.22	1.21	2.736	2.617	94.7	3.15
3.15	393.6	1.19	1.19	1.18	2.745	2.626	94.7	3.11
3.16	392.4	1.17	1.17	1.16	2.755	2.636	94.7	3.06
3.17	391.1	1.14	1.14	1.13	2.764	2.645	94.8	2.99
3.18	389.9	1.11	1.11	1.10	2.773	2.654	94.8	2.92
3.19	388.7	1.08	1.08	1.07	2.782	2.663	94.8	2.85
3.2	387.5	1.06	1.06	1.05	2.792	2.672	94.8	2.80
3.21	386.2	1.03	1.03	1.02	2.801	2.681	94.8	2.75
3.22	385.0	1.01	1.01	1.00	2.810	2.690	94.8	2.69
3.23	383.9	0.99	0.99	0.98	2.819	2.699	94.8	2.64
3.24	382.7	0.97	0.97	0.96	2.829	2.709	94.9	2.61
3.25	381.5	0.95	0.95	0.94	2.838	2.718	94.9	2.56
3.26	380.3	0.92	0.92	0.92	2.847	2.727	94.9	2.49
3.27	379.2	0.90	0.90	0.89	2.856	2.736	94.9	2.44
3.28	378.0	0.87	0.87	0.86	2.865	2.745	94.9	2.37
3.29	376.9	0.84	0.84	0.84	2.874	2.754	94.9	2.30
3.3	375.7	0.82	0.82	0.81	2.883	2.763	94.9	2.24
3.31	374.6	0.80	0.80	0.79	2.893	2.772	95.0	2.19

E_g [eV]	λ_g [nm]	J_{max} [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
3.32	373.4	0.78	0.78	0.77	2.902	2.781	95.0	2.15
3.33	372.3	0.76	0.76	0.75	2.911	2.790	95.0	2.10
3.34	371.2	0.74	0.74	0.73	2.920	2.799	95.0	2.04
3.35	370.1	0.71	0.71	0.71	2.929	2.808	95.0	1.98
3.36	369.0	0.69	0.69	0.68	2.938	2.817	95.0	1.92
3.37	367.9	0.67	0.67	0.66	2.947	2.826	95.0	1.87
3.38	366.8	0.64	0.64	0.64	2.956	2.835	95.0	1.81
3.39	365.7	0.62	0.62	0.62	2.965	2.844	95.1	1.75
3.4	364.7	0.60	0.60	0.59	2.974	2.853	95.1	1.70
3.41	363.6	0.58	0.58	0.58	2.983	2.862	95.1	1.65
3.42	362.5	0.56	0.56	0.56	2.992	2.871	95.1	1.60
3.43	361.5	0.55	0.55	0.54	3.001	2.880	95.1	1.56
3.44	360.4	0.53	0.53	0.53	3.010	2.889	95.1	1.52
3.45	359.4	0.51	0.51	0.51	3.019	2.897	95.1	1.47
3.46	358.3	0.50	0.50	0.49	3.028	2.907	95.1	1.44
3.47	357.3	0.49	0.49	0.48	3.037	2.916	95.2	1.40
3.48	356.3	0.47	0.47	0.47	3.046	2.925	95.2	1.36
3.49	355.3	0.45	0.45	0.45	3.055	2.933	95.2	1.32
3.5	354.2	0.44	0.44	0.43	3.064	2.942	95.2	1.27
3.51	353.2	0.42	0.42	0.42	3.073	2.951	95.2	1.23
3.52	352.2	0.41	0.41	0.40	3.082	2.960	95.2	1.19
3.53	351.2	0.39	0.39	0.39	3.091	2.969	95.2	1.15
3.54	350.2	0.37	0.37	0.37	3.100	2.977	95.2	1.11
3.55	349.3	0.36	0.36	0.36	3.109	2.986	95.2	1.07
3.56	348.3	0.35	0.35	0.34	3.117	2.995	95.3	1.03
3.57	347.3	0.33	0.33	0.33	3.126	3.004	95.3	1.00
3.58	346.3	0.32	0.32	0.32	3.135	3.013	95.3	0.96
3.59	345.4	0.31	0.31	0.31	3.144	3.021	95.3	0.92
3.6	344.4	0.30	0.30	0.29	3.153	3.030	95.3	0.89
3.61	343.4	0.28	0.28	0.28	3.162	3.039	95.3	0.86
3.62	342.5	0.27	0.27	0.27	3.170	3.047	95.3	0.82
3.63	341.6	0.26	0.26	0.26	3.179	3.056	95.3	0.78
3.64	340.6	0.25	0.25	0.24	3.188	3.064	95.3	0.75
3.65	339.7	0.23	0.23	0.23	3.196	3.073	95.4	0.71
3.66	338.8	0.22	0.22	0.22	3.205	3.081	95.4	0.68
3.67	337.8	0.21	0.21	0.21	3.213	3.090	95.4	0.65
3.68	336.9	0.20	0.20	0.20	3.222	3.098	95.4	0.62
3.69	336.0	0.19	0.19	0.19	3.230	3.107	95.4	0.59
3.7	335.1	0.18	0.18	0.18	3.239	3.115	95.4	0.56
3.71	334.2	0.17	0.17	0.17	3.247	3.123	95.4	0.52
3.72	333.3	0.16	0.16	0.16	3.255	3.132	95.4	0.49
3.73	332.4	0.15	0.15	0.15	3.263	3.140	95.4	0.46
3.74	331.5	0.14	0.14	0.14	3.271	3.148	95.4	0.43
3.75	330.6	0.13	0.13	0.13	3.279	3.156	95.4	0.40
3.76	329.7	0.12	0.12	0.12	3.287	3.163	95.5	0.37

E_g [eV]	λ_g [nm]	J_{max} [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
3.77	328.9	0.11	0.11	0.11	3.295	3.171	95.5	0.34
3.78	328.0	0.10	0.10	0.10	3.302	3.178	95.5	0.31
3.79	327.1	9.08E-02	9.08E-02	9.01E-02	3.310	3.186	95.5	0.29
3.8	326.3	8.18E-02	8.18E-02	8.11E-02	3.317	3.193	95.5	0.26
3.81	325.4	7.36E-02	7.36E-02	7.30E-02	3.324	3.200	95.5	0.23
3.82	324.6	6.71E-02	6.71E-02	6.66E-02	3.332	3.207	95.5	0.21
3.83	323.7	6.11E-02	6.11E-02	6.06E-02	3.339	3.215	95.5	0.19
3.84	322.9	5.59E-02	5.59E-02	5.54E-02	3.347	3.222	95.5	0.18
3.85	322.0	5.11E-02	5.11E-02	5.07E-02	3.354	3.230	95.5	0.16
3.86	321.2	4.60E-02	4.60E-02	4.57E-02	3.361	3.237	95.5	0.15
3.87	320.4	4.08E-02	4.08E-02	4.05E-02	3.368	3.244	95.5	0.13
3.88	319.5	3.63E-02	3.63E-02	3.60E-02	3.375	3.250	95.6	0.12
3.89	318.7	3.21E-02	3.21E-02	3.19E-02	3.382	3.257	95.6	0.10
3.9	317.9	2.82E-02	2.82E-02	2.80E-02	3.388	3.264	95.6	9.14E-02
3.91	317.1	2.45E-02	2.45E-02	2.43E-02	3.395	3.270	95.6	7.95E-02
3.92	316.3	2.13E-02	2.13E-02	2.11E-02	3.401	3.276	95.6	6.92E-02
3.93	315.5	1.87E-02	1.87E-02	1.85E-02	3.407	3.283	95.6	6.08E-02
3.94	314.7	1.60E-02	1.60E-02	1.59E-02	3.413	3.288	95.6	5.23E-02
3.95	313.9	1.35E-02	1.35E-02	1.34E-02	3.419	3.294	95.6	4.42E-02
3.96	313.1	1.13E-02	1.13E-02	1.12E-02	3.424	3.299	95.6	3.70E-02
3.97	312.3	9.28E-03	9.28E-03	9.20E-03	3.429	3.304	95.6	3.04E-02
3.98	311.5	7.47E-03	7.47E-03	7.41E-03	3.433	3.308	95.6	2.45E-02
3.99	310.7	5.89E-03	5.89E-03	5.85E-03	3.437	3.312	95.6	1.94E-02
4	310.0	4.67E-03	4.67E-03	4.64E-03	3.441	3.316	95.6	1.54E-02
4.01	309.2	3.80E-03	3.80E-03	3.78E-03	3.445	3.320	95.6	1.25E-02
4.02	308.4	3.02E-03	3.02E-03	2.99E-03	3.449	3.324	95.6	9.95E-03
4.03	307.7	2.29E-03	2.29E-03	2.27E-03	3.452	3.327	95.6	7.55E-03
4.04	306.9	1.69E-03	1.69E-03	1.68E-03	3.454	3.329	95.6	5.58E-03
4.05	306.1	1.27E-03	1.27E-03	1.26E-03	3.457	3.331	95.6	4.21E-03
4.06	305.4	9.26E-04	9.26E-04	9.19E-04	3.458	3.333	95.6	3.06E-03
4.07	304.6	6.32E-04	6.32E-04	6.28E-04	3.458	3.333	95.6	2.09E-03
4.08	303.9	4.29E-04	4.29E-04	4.26E-04	3.458	3.333	95.6	1.42E-03
4.09	303.1	2.72E-04	2.72E-04	2.70E-04	3.456	3.331	95.6	8.98E-04
4.1	302.4	1.66E-04	1.66E-04	1.65E-04	3.454	3.328	95.6	5.49E-04
4.11	301.7	1.08E-04	1.08E-04	1.07E-04	3.452	3.327	95.6	3.57E-04
4.12	300.9	6.61E-05	6.61E-05	6.56E-05	3.450	3.325	95.6	2.18E-04
4.13	300.2	4.14E-05	4.14E-05	4.10E-05	3.447	3.322	95.6	1.36E-04
4.14	299.5	2.40E-05	2.40E-05	2.39E-05	3.444	3.318	95.6	7.92E-05
4.15	298.8	1.37E-05	1.37E-05	1.36E-05	3.439	3.314	95.6	4.52E-05
4.16	298.0	7.52E-06	7.52E-06	7.46E-06	3.433	3.308	95.6	2.47E-05
4.17	297.3	3.76E-06	3.76E-06	3.73E-06	3.425	3.300	95.6	1.23E-05
4.18	296.6	1.88E-06	1.88E-06	1.86E-06	3.417	3.292	95.6	6.14E-06
4.19	295.9	8.53E-07	8.53E-07	8.46E-07	3.406	3.282	95.6	2.78E-06
4.2	295.2	3.44E-07	3.44E-07	3.41E-07	3.392	3.268	95.6	1.12E-06
4.21	294.5	1.36E-07	1.36E-07	1.35E-07	3.380	3.255	95.6	4.38E-07

E_g [eV]	λ_g [nm]	J_{max} [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
4.22	293.8	5.68E-08	5.68E-08	5.64E-08	3.366	3.241	95.5	1.83E-07
4.23	293.1	1.92E-08	1.92E-08	1.91E-08	3.348	3.223	95.5	6.15E-08
4.24	292.4	6.95E-09	6.95E-09	6.89E-09	3.332	3.208	95.5	2.21E-08
4.25	291.7	2.33E-09	2.33E-09	2.31E-09	3.312	3.188	95.5	7.37E-09
4.26	291.0	5.44E-10	5.44E-10	5.39E-10	3.286	3.162	95.5	1.71E-09
4.27	290.4	1.53E-10	1.53E-10	1.51E-10	3.262	3.139	95.4	4.75E-10
4.28	289.7	3.89E-11	3.89E-11	3.85E-11	3.235	3.112	95.4	1.20E-10
4.29	289.0	6.40E-12	6.40E-12	6.35E-12	3.202	3.079	95.4	1.95E-11
4.3	288.3	1.44E-12	1.44E-12	1.43E-12	3.171	3.048	95.3	4.37E-12
4.31	287.7	2.48E-13	2.48E-13	2.46E-13	3.131	3.009	95.3	7.44E-13
4.32	287.0	2.78E-14	2.78E-14	2.76E-14	3.093	2.970	95.2	8.18E-14
4.33	286.3	5.21E-15	5.21E-15	5.17E-15	3.054	2.932	95.2	1.52E-14
4.34	285.7	7.43E-16	7.43E-16	7.37E-16	3.014	2.892	95.1	2.14E-15
4.35	285.0	9.59E-17	9.59E-17	9.51E-17	2.975	2.854	95.1	2.72E-16
4.36	284.4	1.84E-17	1.84E-17	1.82E-17	2.939	2.819	95.0	5.15E-17
4.37	283.7	2.37E-18	2.37E-18	2.35E-18	2.886	2.766	94.9	6.56E-18
4.38	283.1	2.15E-19	2.15E-19	2.13E-19	2.848	2.727	94.9	5.81E-19
4.39	282.4	3.64E-20	3.64E-20	3.60E-20	2.811	2.691	94.8	9.71E-20
4.4	281.8	5.31E-21	5.31E-21	5.26E-21	2.762	2.643	94.8	1.40E-20
4.41	281.1	3.11E-22	3.11E-22	3.08E-22	2.685	2.566	94.6	8.03E-22
4.42	280.5	8.06E-24	8.06E-24	7.98E-24	2.626	2.508	94.5	2.00E-23

Table S2. Detailed balance limit for a single junction solar cell at a cell temperature T_c of 25 °C illuminated by the AM 1.5G spectral irradiance (ASTM G173-03) as a function of the energy band gap E_g and band gap wavelength λ_g considering radiative emission only from the front side due to a perfect reflector at the rear side as shown in Fig. 2b.

E_g [eV]	λ_g [nm]	$J_{\max}$ [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
0.31	3999.5	68.98	68.72	55.08	0.143	0.102	57.1	5.60
0.32	3874.5	68.69	68.50	55.57	0.151	0.109	58.3	6.04
0.33	3757.1	68.38	68.24	55.96	0.160	0.116	59.5	6.48
0.34	3646.6	68.06	67.96	56.31	0.168	0.123	60.6	6.92
0.35	3542.4	67.76	67.69	56.60	0.177	0.130	61.7	7.37
0.36	3444.0	67.45	67.40	56.84	0.185	0.138	62.7	7.82
0.37	3350.9	67.25	67.21	57.14	0.194	0.145	63.7	8.29
0.38	3262.7	67.15	67.13	57.48	0.202	0.153	64.6	8.77
0.39	3179.1	67.08	67.07	57.83	0.211	0.160	65.5	9.27
0.4	3099.6	66.95	66.94	58.06	0.220	0.168	66.3	9.76
0.41	3024.0	66.89	66.88	58.35	0.229	0.176	67.1	10.26
0.42	2952.0	66.81	66.81	58.62	0.238	0.184	67.8	10.76
0.43	2883.4	66.77	66.77	58.87	0.246	0.192	68.6	11.27
0.44	2817.8	66.77	66.77	59.16	0.255	0.199	69.2	11.80
0.45	2755.2	66.77	66.77	59.41	0.264	0.208	69.9	12.33
0.46	2695.3	66.77	66.77	59.67	0.273	0.216	70.5	12.86
0.47	2638.0	66.77	66.77	59.90	0.282	0.224	71.1	13.39
0.48	2583.0	66.77	66.77	60.11	0.291	0.232	71.7	13.93
0.49	2530.3	66.77	66.77	60.32	0.300	0.240	72.3	14.47
0.5	2479.7	66.75	66.74	60.48	0.309	0.248	72.8	15.01
0.51	2431.1	66.52	66.52	60.47	0.318	0.256	73.3	15.50
0.52	2384.3	66.20	66.20	60.35	0.327	0.265	73.8	15.96
0.53	2339.3	65.83	65.83	60.16	0.336	0.273	74.2	16.41
0.54	2296.0	65.36	65.36	59.89	0.345	0.281	74.7	16.82
0.55	2254.3	64.86	64.86	59.57	0.354	0.289	75.1	17.23
0.56	2214.0	64.32	64.32	59.21	0.363	0.298	75.5	17.61
0.57	2175.2	63.79	63.79	58.86	0.371	0.306	75.9	17.99
0.58	2137.7	63.24	63.24	58.46	0.380	0.314	76.3	18.35
0.59	2101.4	62.69	62.69	58.06	0.389	0.322	76.7	18.71
0.6	2066.4	62.20	62.20	57.72	0.398	0.331	77.1	19.08
0.61	2032.5	61.78	61.78	57.43	0.407	0.339	77.4	19.47
0.62	1999.7	61.53	61.53	57.30	0.416	0.348	77.7	19.91
0.63	1968.0	61.17	61.17	57.05	0.425	0.356	78.1	20.31
0.64	1937.3	61.09	61.09	57.07	0.435	0.365	78.4	20.81
0.65	1907.4	61.08	61.08	57.15	0.444	0.373	78.7	21.33
0.66	1878.5	61.08	61.08	57.23	0.453	0.382	79.0	21.86
0.67	1850.5	61.08	61.08	57.31	0.462	0.391	79.3	22.39
0.68	1823.3	61.08	61.08	57.39	0.472	0.400	79.6	22.92
0.69	1796.9	61.03	61.03	57.42	0.481	0.408	79.9	23.44
0.7	1771.2	60.70	60.70	57.18	0.490	0.417	80.1	23.83

E_g [eV]	λ_g [nm]	$J_{\max}$ [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
0.71	1746.3	60.17	60.17	56.75	0.499	0.425	80.4	24.14
0.72	1722.0	59.60	59.60	56.27	0.508	0.434	80.6	24.42
0.73	1698.4	58.97	58.97	55.73	0.517	0.443	80.9	24.66
0.74	1675.5	58.33	58.33	55.19	0.526	0.451	81.1	24.89
0.75	1653.1	57.68	57.68	54.62	0.535	0.460	81.4	25.11
0.76	1631.4	57.03	57.03	54.07	0.544	0.468	81.6	25.32
0.77	1610.2	56.38	56.38	53.50	0.553	0.477	81.8	25.51
0.78	1589.5	55.74	55.74	52.94	0.563	0.486	82.0	25.70
0.79	1569.4	55.12	55.12	52.39	0.572	0.494	82.2	25.89
0.8	1549.8	54.46	54.46	51.81	0.581	0.503	82.4	26.05
0.81	1530.7	53.81	53.81	51.24	0.590	0.512	82.6	26.21
0.82	1512.0	53.22	53.22	50.71	0.599	0.520	82.8	26.38
0.83	1493.8	52.73	52.73	50.29	0.608	0.529	83.0	26.60
0.84	1476.0	52.47	52.47	50.08	0.617	0.538	83.2	26.93
0.85	1458.6	52.29	52.29	49.94	0.627	0.547	83.4	27.30
0.86	1441.7	52.15	52.15	49.84	0.636	0.556	83.5	27.69
0.87	1425.1	52.09	52.09	49.82	0.645	0.565	83.7	28.13
0.88	1408.9	52.06	52.06	49.83	0.655	0.574	83.9	28.58
0.89	1393.1	52.06	52.06	49.87	0.664	0.583	84.1	29.05
0.9	1377.6	52.06	52.06	49.89	0.674	0.592	84.2	29.52
0.91	1362.5	52.06	52.06	49.93	0.683	0.601	84.4	30.00
0.92	1347.7	52.06	52.06	49.95	0.693	0.610	84.5	30.46
0.93	1333.2	51.83	51.83	49.76	0.702	0.619	84.7	30.80
0.94	1319.0	51.46	51.46	49.44	0.711	0.628	84.8	31.04
0.95	1305.1	50.98	50.98	49.01	0.720	0.637	85.0	31.20
0.96	1291.5	50.44	50.44	48.51	0.730	0.646	85.1	31.31
0.97	1278.2	49.87	49.87	47.99	0.739	0.655	85.3	31.40
0.98	1265.1	49.35	49.35	47.51	0.748	0.663	85.4	31.51
0.99	1252.4	48.80	48.80	47.00	0.757	0.672	85.5	31.59
1	1239.8	48.22	48.22	46.47	0.766	0.681	85.7	31.64
1.01	1227.6	47.65	47.65	45.94	0.776	0.690	85.8	31.69
1.02	1215.5	47.11	47.11	45.44	0.785	0.699	85.9	31.75
1.03	1203.7	46.61	46.61	44.98	0.794	0.708	86.0	31.83
1.04	1192.2	46.12	46.12	44.53	0.803	0.717	86.2	31.91
1.05	1180.8	45.66	45.66	44.10	0.813	0.726	86.3	32.00
1.06	1169.7	45.20	45.20	43.67	0.822	0.735	86.4	32.08
1.07	1158.7	44.80	44.80	43.31	0.831	0.744	86.5	32.20
1.08	1148.0	44.56	44.56	43.09	0.841	0.753	86.6	32.43
1.09	1137.5	44.36	44.36	42.92	0.850	0.762	86.7	32.69
1.1	1127.1	44.23	44.23	42.80	0.859	0.771	86.8	33.00
1.11	1117.0	44.11	44.11	42.70	0.869	0.780	86.9	33.31
1.12	1107.0	43.81	43.81	42.43	0.878	0.789	87.1	33.48
1.13	1097.2	43.39	43.39	42.04	0.888	0.798	87.2	33.55
1.14	1087.6	42.92	42.92	41.60	0.897	0.807	87.3	33.58
1.15	1078.1	42.44	42.44	41.14	0.906	0.816	87.4	33.58

E_g [eV]	λ_g [nm]	J_{max} [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
1.16	1068.8	41.95	41.95	40.68	0.915	0.825	87.5	33.57
1.17	1059.7	41.46	41.46	40.22	0.925	0.834	87.5	33.55
1.18	1050.7	40.97	40.97	39.76	0.934	0.843	87.6	33.52
1.19	1041.9	40.48	40.48	39.29	0.943	0.852	87.7	33.48
1.2	1033.2	39.99	39.99	38.83	0.952	0.861	87.8	33.44
1.21	1024.7	39.50	39.50	38.36	0.962	0.870	87.9	33.38
1.22	1016.3	39.01	39.01	37.91	0.971	0.879	88.0	33.32
1.23	1008.0	38.53	38.53	37.45	0.980	0.888	88.1	33.26
1.24	999.9	38.05	38.05	36.99	0.990	0.898	88.2	33.19
1.25	991.9	37.57	37.57	36.54	0.999	0.907	88.3	33.11
1.26	984.0	37.11	37.11	36.10	1.008	0.916	88.3	33.04
1.27	976.3	36.71	36.71	35.72	1.017	0.925	88.4	33.02
1.28	968.6	36.33	36.33	35.36	1.027	0.934	88.5	33.00
1.29	961.1	36.03	36.03	35.08	1.036	0.943	88.6	33.06
1.3	953.7	35.82	35.82	34.88	1.046	0.952	88.7	33.19
1.31	946.4	35.63	35.63	34.70	1.055	0.961	88.7	33.35
1.32	939.3	35.44	35.44	34.53	1.065	0.971	88.8	33.50
1.33	932.2	35.32	35.32	34.42	1.074	0.980	88.9	33.71
1.34	925.3	35.03	35.03	34.15	1.083	0.989	89.0	33.77
1.35	918.4	34.66	34.66	33.79	1.093	0.998	89.0	33.72
1.36	911.6	34.34	34.34	33.48	1.102	1.007	89.1	33.72
1.37	905.0	34.00	34.00	33.16	1.112	1.016	89.2	33.70
1.38	898.4	33.67	33.67	32.85	1.121	1.026	89.3	33.67
1.39	892.0	33.30	33.30	32.50	1.130	1.035	89.3	33.61
1.4	885.6	32.88	32.88	32.09	1.140	1.044	89.4	33.49
1.41	879.3	32.47	32.47	31.69	1.149	1.053	89.5	33.36
1.42	873.1	32.05	32.05	31.29	1.158	1.062	89.5	33.22
1.43	867.0	31.64	31.64	30.90	1.168	1.071	89.6	33.09
1.44	861.0	31.24	31.24	30.52	1.177	1.080	89.7	32.95
1.45	855.1	30.84	30.84	30.13	1.186	1.089	89.7	32.81
1.46	849.2	30.46	30.46	29.77	1.196	1.098	89.8	32.68
1.47	843.4	30.07	30.07	29.39	1.205	1.108	89.8	32.54
1.48	837.7	29.68	29.68	29.01	1.214	1.117	89.9	32.39
1.49	832.1	29.31	29.31	28.66	1.224	1.126	90.0	32.25
1.5	826.6	28.97	28.97	28.33	1.233	1.135	90.0	32.14
1.51	821.1	28.64	28.64	28.01	1.242	1.144	90.1	32.04
1.52	815.7	28.33	28.33	27.72	1.252	1.153	90.1	31.95
1.53	810.4	27.99	27.99	27.39	1.261	1.163	90.2	31.83
1.54	805.1	27.63	27.63	27.03	1.270	1.172	90.3	31.66
1.55	799.9	27.26	27.26	26.68	1.280	1.181	90.3	31.50
1.56	794.8	26.90	26.90	26.34	1.289	1.190	90.4	31.33
1.57	789.7	26.55	26.55	25.99	1.298	1.199	90.4	31.16
1.58	784.7	26.19	26.19	25.64	1.308	1.208	90.5	30.97
1.59	779.8	25.83	25.83	25.29	1.317	1.217	90.5	30.78
1.6	774.9	25.47	25.47	24.94	1.326	1.226	90.6	30.58

E_g [eV]	λ_g [nm]	$J_{\max}$ [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
1.61	770.1	25.12	25.12	24.60	1.336	1.236	90.6	30.39
1.62	765.3	24.82	24.82	24.32	1.345	1.245	90.7	30.26
1.63	760.6	24.67	24.67	24.18	1.355	1.254	90.7	30.31
1.64	756.0	24.40	24.40	23.91	1.364	1.263	90.8	30.20
1.65	751.4	24.05	24.05	23.58	1.373	1.273	90.8	29.99
1.66	746.9	23.71	23.71	23.25	1.383	1.282	90.9	29.79
1.67	742.4	23.38	23.38	22.92	1.392	1.291	90.9	29.58
1.68	738.0	23.06	23.06	22.61	1.401	1.300	91.0	29.39
1.69	733.6	22.74	22.74	22.31	1.411	1.309	91.0	29.19
1.7	729.3	22.46	22.46	22.03	1.420	1.318	91.1	29.03
1.71	725.1	22.19	22.19	21.77	1.429	1.328	91.1	28.89
1.72	720.8	21.91	21.91	21.50	1.439	1.337	91.2	28.73
1.73	716.7	21.67	21.67	21.26	1.448	1.346	91.2	28.61
1.74	712.6	21.37	21.37	20.97	1.458	1.355	91.2	28.41
1.75	708.5	21.06	21.06	20.67	1.467	1.364	91.3	28.19
1.76	704.5	20.76	20.76	20.38	1.476	1.374	91.3	27.98
1.77	700.5	20.47	20.47	20.10	1.486	1.383	91.4	27.78
1.78	696.5	20.18	20.18	19.82	1.495	1.392	91.4	27.57
1.79	692.6	19.91	19.91	19.55	1.504	1.401	91.5	27.38
1.8	688.8	19.65	19.65	19.29	1.514	1.410	91.5	27.20
1.81	685.0	19.40	19.40	19.05	1.523	1.420	91.5	27.04
1.82	681.2	19.11	19.11	18.77	1.532	1.429	91.6	26.81
1.83	677.5	18.83	18.83	18.50	1.542	1.438	91.6	26.59
1.84	673.8	18.54	18.54	18.22	1.551	1.447	91.7	26.36
1.85	670.2	18.27	18.27	17.95	1.560	1.456	91.7	26.13
1.86	666.6	17.99	17.99	17.68	1.570	1.465	91.7	25.90
1.87	663.0	17.72	17.72	17.42	1.579	1.475	91.8	25.68
1.88	659.5	17.46	17.46	17.16	1.588	1.484	91.8	25.46
1.89	656.0	17.22	17.22	16.92	1.598	1.493	91.9	25.26
1.9	652.5	16.97	16.97	16.68	1.607	1.502	91.9	25.05
1.91	649.1	16.72	16.72	16.44	1.617	1.511	91.9	24.84
1.92	645.8	16.47	16.47	16.20	1.626	1.521	92.0	24.62
1.93	642.4	16.22	16.22	15.95	1.635	1.530	92.0	24.39
1.94	639.1	15.97	15.97	15.71	1.645	1.539	92.0	24.17
1.95	635.8	15.73	15.73	15.47	1.654	1.548	92.1	23.94
1.96	632.6	15.49	15.49	15.24	1.663	1.557	92.1	23.72
1.97	629.4	15.26	15.26	15.02	1.673	1.567	92.1	23.51
1.98	626.2	15.04	15.04	14.79	1.682	1.576	92.2	23.30
1.99	623.0	14.81	14.81	14.58	1.691	1.585	92.2	23.09
2	619.9	14.59	14.59	14.35	1.701	1.594	92.3	22.88
2.01	616.8	14.36	14.36	14.13	1.710	1.603	92.3	22.66
2.02	613.8	14.14	14.14	13.92	1.719	1.613	92.3	22.44
2.03	610.8	13.92	13.92	13.71	1.729	1.622	92.4	22.22
2.04	607.8	13.71	13.71	13.50	1.738	1.631	92.4	22.00
2.05	604.8	13.49	13.49	13.28	1.747	1.640	92.4	21.78

E_g [eV]	λ_g [nm]	$J_{\max}$ [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
2.06	601.9	13.28	13.28	13.08	1.757	1.649	92.5	21.56
2.07	599.0	13.08	13.08	12.88	1.766	1.659	92.5	21.35
2.08	596.1	12.87	12.87	12.68	1.775	1.668	92.5	21.14
2.09	593.2	12.67	12.67	12.48	1.785	1.677	92.5	20.93
2.1	590.4	12.48	12.48	12.29	1.794	1.686	92.6	20.72
2.11	587.6	12.29	12.29	12.11	1.804	1.696	92.6	20.53
2.12	584.8	12.10	12.10	11.92	1.813	1.705	92.6	20.31
2.13	582.1	11.90	11.90	11.72	1.822	1.714	92.7	20.08
2.14	579.4	11.71	11.71	11.53	1.832	1.723	92.7	19.87
2.15	576.7	11.52	11.52	11.35	1.841	1.732	92.7	19.66
2.16	574.0	11.34	11.34	11.17	1.850	1.742	92.8	19.45
2.17	571.4	11.15	11.15	10.99	1.860	1.751	92.8	19.24
2.18	568.7	10.97	10.97	10.82	1.869	1.760	92.8	19.03
2.19	566.1	10.80	10.80	10.64	1.878	1.769	92.8	18.82
2.2	563.6	10.62	10.62	10.47	1.888	1.778	92.9	18.61
2.21	561.0	10.44	10.44	10.30	1.897	1.788	92.9	18.40
2.22	558.5	10.27	10.27	10.13	1.906	1.797	92.9	18.19
2.23	556.0	10.10	10.10	9.96	1.916	1.806	93.0	17.99
2.24	553.5	9.93	9.93	9.79	1.925	1.815	93.0	17.77
2.25	551.0	9.76	9.76	9.63	1.934	1.825	93.0	17.56
2.26	548.6	9.60	9.60	9.46	1.944	1.834	93.0	17.35
2.27	546.2	9.43	9.43	9.30	1.953	1.843	93.1	17.14
2.28	543.8	9.27	9.27	9.14	1.962	1.852	93.1	16.93
2.29	541.4	9.11	9.11	8.99	1.972	1.861	93.1	16.72
2.3	539.1	8.96	8.96	8.84	1.981	1.871	93.1	16.52
2.31	536.7	8.80	8.80	8.68	1.990	1.880	93.2	16.31
2.32	534.4	8.64	8.64	8.53	2.000	1.889	93.2	16.10
2.33	532.1	8.50	8.50	8.38	2.009	1.898	93.2	15.91
2.34	529.8	8.34	8.34	8.23	2.018	1.907	93.2	15.69
2.35	527.6	8.19	8.19	8.08	2.028	1.917	93.3	15.49
2.36	525.4	8.05	8.05	7.95	2.037	1.926	93.3	15.30
2.37	523.1	7.91	7.91	7.80	2.046	1.935	93.3	15.09
2.38	520.9	7.76	7.76	7.66	2.056	1.944	93.3	14.89
2.39	518.8	7.63	7.63	7.53	2.065	1.953	93.4	14.70
2.4	516.6	7.50	7.50	7.41	2.074	1.963	93.4	14.53
2.41	514.5	7.37	7.37	7.28	2.084	1.972	93.4	14.34
2.42	512.3	7.24	7.24	7.14	2.093	1.981	93.4	14.15
2.43	510.2	7.10	7.10	7.01	2.102	1.990	93.5	13.94
2.44	508.1	6.96	6.96	6.88	2.112	1.999	93.5	13.74
2.45	506.1	6.83	6.83	6.75	2.121	2.009	93.5	13.54
2.46	504.0	6.70	6.70	6.62	2.130	2.018	93.5	13.35
2.47	502.0	6.58	6.58	6.49	2.139	2.027	93.6	13.16
2.48	499.9	6.45	6.45	6.37	2.149	2.036	93.6	12.97
2.49	497.9	6.33	6.33	6.25	2.158	2.045	93.6	12.77
2.5	495.9	6.20	6.20	6.12	2.167	2.054	93.6	12.58

E_g [eV]	λ_g [nm]	J_{max} [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
2.51	494.0	6.07	6.07	6.00	2.177	2.064	93.6	12.38
2.52	492.0	5.95	5.95	5.88	2.186	2.073	93.7	12.18
2.53	490.1	5.83	5.83	5.76	2.195	2.082	93.7	11.99
2.54	488.1	5.72	5.72	5.65	2.204	2.091	93.7	11.81
2.55	486.2	5.61	5.61	5.54	2.214	2.100	93.7	11.64
2.56	484.3	5.50	5.50	5.44	2.223	2.109	93.7	11.46
2.57	482.4	5.38	5.38	5.32	2.232	2.119	93.8	11.27
2.58	480.6	5.27	5.27	5.20	2.242	2.128	93.8	11.07
2.59	478.7	5.15	5.15	5.09	2.251	2.137	93.8	10.87
2.6	476.9	5.04	5.04	4.98	2.260	2.146	93.8	10.68
2.61	475.0	4.93	4.93	4.87	2.269	2.155	93.9	10.49
2.62	473.2	4.82	4.82	4.76	2.278	2.164	93.9	10.30
2.63	471.4	4.71	4.71	4.65	2.288	2.173	93.9	10.11
2.64	469.6	4.61	4.61	4.55	2.297	2.182	93.9	9.93
2.65	467.9	4.50	4.50	4.45	2.306	2.192	93.9	9.75
2.66	466.1	4.40	4.40	4.35	2.315	2.201	94.0	9.57
2.67	464.4	4.30	4.30	4.25	2.325	2.210	94.0	9.39
2.68	462.6	4.20	4.20	4.15	2.334	2.219	94.0	9.20
2.69	460.9	4.09	4.09	4.05	2.343	2.228	94.0	9.01
2.7	459.2	4.00	4.00	3.95	2.352	2.237	94.0	8.84
2.71	457.5	3.90	3.90	3.86	2.361	2.246	94.0	8.66
2.72	455.8	3.80	3.80	3.76	2.370	2.255	94.1	8.47
2.73	454.2	3.71	3.71	3.67	2.380	2.264	94.1	8.30
2.74	452.5	3.62	3.62	3.58	2.389	2.273	94.1	8.13
2.75	450.9	3.52	3.52	3.49	2.398	2.282	94.1	7.95
2.76	449.2	3.43	3.43	3.39	2.407	2.291	94.1	7.77
2.77	447.6	3.34	3.34	3.31	2.416	2.300	94.2	7.60
2.78	446.0	3.26	3.26	3.22	2.425	2.309	94.2	7.44
2.79	444.4	3.18	3.18	3.14	2.435	2.319	94.2	7.29
2.8	442.8	3.10	3.10	3.06	2.444	2.328	94.2	7.13
2.81	441.2	3.02	3.02	2.99	2.453	2.337	94.2	6.98
2.82	439.7	2.95	2.95	2.91	2.462	2.346	94.2	6.83
2.83	438.1	2.88	2.88	2.85	2.471	2.355	94.3	6.70
2.84	436.6	2.81	2.81	2.78	2.480	2.364	94.3	6.56
2.85	435.0	2.73	2.73	2.70	2.490	2.373	94.3	6.42
2.86	433.5	2.67	2.67	2.64	2.499	2.382	94.3	6.29
2.87	432.0	2.60	2.60	2.58	2.508	2.391	94.3	6.16
2.88	430.5	2.55	2.55	2.53	2.517	2.400	94.3	6.06
2.89	429.0	2.51	2.51	2.48	2.527	2.410	94.4	5.97
2.9	427.5	2.45	2.45	2.42	2.536	2.419	94.4	5.85
2.91	426.1	2.39	2.39	2.36	2.545	2.428	94.4	5.73
2.92	424.6	2.33	2.33	2.30	2.554	2.437	94.4	5.60
2.93	423.2	2.26	2.26	2.24	2.563	2.446	94.4	5.48
2.94	421.7	2.20	2.20	2.18	2.572	2.455	94.4	5.35
2.95	420.3	2.14	2.14	2.12	2.582	2.464	94.5	5.23

E_g [eV]	λ_g [nm]	J_{max} [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
2.96	418.9	2.09	2.09	2.07	2.591	2.473	94.5	5.11
2.97	417.5	2.03	2.03	2.01	2.600	2.482	94.5	4.98
2.98	416.1	1.97	1.97	1.95	2.609	2.491	94.5	4.86
2.99	414.7	1.91	1.91	1.89	2.618	2.500	94.5	4.74
3	413.3	1.86	1.86	1.84	2.627	2.509	94.5	4.62
3.01	411.9	1.80	1.80	1.79	2.636	2.518	94.6	4.49
3.02	410.5	1.75	1.75	1.73	2.645	2.527	94.6	4.38
3.03	409.2	1.70	1.70	1.68	2.654	2.536	94.6	4.27
3.04	407.8	1.65	1.65	1.63	2.663	2.545	94.6	4.15
3.05	406.5	1.60	1.60	1.58	2.672	2.554	94.6	4.04
3.06	405.2	1.55	1.55	1.53	2.681	2.563	94.6	3.93
3.07	403.9	1.50	1.50	1.48	2.690	2.572	94.6	3.82
3.08	402.5	1.45	1.45	1.43	2.699	2.581	94.7	3.70
3.09	401.2	1.40	1.40	1.39	2.708	2.590	94.7	3.59
3.1	399.9	1.35	1.35	1.34	2.717	2.598	94.7	3.48
3.11	398.7	1.31	1.31	1.29	2.726	2.607	94.7	3.37
3.12	397.4	1.27	1.27	1.26	2.735	2.616	94.7	3.29
3.13	396.1	1.25	1.25	1.24	2.745	2.626	94.7	3.25
3.14	394.9	1.22	1.22	1.21	2.754	2.635	94.7	3.18
3.15	393.6	1.19	1.19	1.18	2.763	2.644	94.8	3.13
3.16	392.4	1.17	1.17	1.16	2.773	2.653	94.8	3.08
3.17	391.1	1.14	1.14	1.13	2.782	2.662	94.8	3.01
3.18	389.9	1.11	1.11	1.10	2.791	2.671	94.8	2.94
3.19	388.7	1.08	1.08	1.07	2.800	2.680	94.8	2.87
3.2	387.5	1.06	1.06	1.05	2.809	2.690	94.8	2.82
3.21	386.2	1.03	1.03	1.02	2.819	2.699	94.8	2.76
3.22	385.0	1.01	1.01	1.00	2.828	2.708	94.9	2.71
3.23	383.9	0.99	0.99	0.98	2.837	2.717	94.9	2.66
3.24	382.7	0.97	0.97	0.96	2.847	2.726	94.9	2.62
3.25	381.5	0.95	0.95	0.94	2.856	2.736	94.9	2.57
3.26	380.3	0.92	0.92	0.92	2.865	2.745	94.9	2.51
3.27	379.2	0.90	0.90	0.89	2.874	2.754	94.9	2.45
3.28	378.0	0.87	0.87	0.86	2.883	2.763	94.9	2.38
3.29	376.9	0.84	0.84	0.84	2.892	2.772	95.0	2.32
3.3	375.7	0.82	0.82	0.81	2.901	2.781	95.0	2.26
3.31	374.6	0.80	0.80	0.79	2.910	2.790	95.0	2.21
3.32	373.4	0.78	0.78	0.77	2.920	2.799	95.0	2.16
3.33	372.3	0.76	0.76	0.75	2.929	2.808	95.0	2.11
3.34	371.2	0.74	0.74	0.73	2.938	2.817	95.0	2.05
3.35	370.1	0.71	0.71	0.71	2.947	2.826	95.0	2.00
3.36	369.0	0.69	0.69	0.68	2.956	2.835	95.0	1.93
3.37	367.9	0.67	0.67	0.66	2.965	2.844	95.1	1.88
3.38	366.8	0.64	0.64	0.64	2.974	2.853	95.1	1.82
3.39	365.7	0.62	0.62	0.62	2.983	2.861	95.1	1.76
3.4	364.7	0.60	0.60	0.59	2.992	2.870	95.1	1.71

E_g [eV]	λ_g [nm]	J_{max} [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
3.41	363.6	0.58	0.58	0.58	3.001	2.879	95.1	1.66
3.42	362.5	0.56	0.56	0.56	3.010	2.888	95.1	1.61
3.43	361.5	0.55	0.55	0.54	3.019	2.897	95.1	1.57
3.44	360.4	0.53	0.53	0.53	3.028	2.906	95.1	1.53
3.45	359.4	0.51	0.51	0.51	3.037	2.915	95.2	1.48
3.46	358.3	0.50	0.50	0.49	3.046	2.924	95.2	1.44
3.47	357.3	0.49	0.49	0.48	3.055	2.933	95.2	1.41
3.48	356.3	0.47	0.47	0.47	3.064	2.942	95.2	1.37
3.49	355.3	0.45	0.45	0.45	3.073	2.951	95.2	1.33
3.5	354.2	0.44	0.44	0.43	3.082	2.960	95.2	1.28
3.51	353.2	0.42	0.42	0.42	3.091	2.969	95.2	1.24
3.52	352.2	0.41	0.41	0.40	3.100	2.977	95.2	1.20
3.53	351.2	0.39	0.39	0.39	3.109	2.986	95.2	1.16
3.54	350.2	0.37	0.37	0.37	3.118	2.995	95.3	1.11
3.55	349.3	0.36	0.36	0.36	3.126	3.004	95.3	1.07
3.56	348.3	0.35	0.35	0.34	3.135	3.013	95.3	1.04
3.57	347.3	0.33	0.33	0.33	3.144	3.021	95.3	1.00
3.58	346.3	0.32	0.32	0.32	3.153	3.030	95.3	0.97
3.59	345.4	0.31	0.31	0.31	3.162	3.039	95.3	0.93
3.6	344.4	0.30	0.30	0.29	3.171	3.048	95.3	0.90
3.61	343.4	0.28	0.28	0.28	3.179	3.056	95.3	0.86
3.62	342.5	0.27	0.27	0.27	3.188	3.065	95.3	0.83
3.63	341.6	0.26	0.26	0.26	3.197	3.074	95.4	0.79
3.64	340.6	0.25	0.25	0.24	3.205	3.082	95.4	0.75
3.65	339.7	0.23	0.23	0.23	3.214	3.091	95.4	0.72
3.66	338.8	0.22	0.22	0.22	3.222	3.099	95.4	0.68
3.67	337.8	0.21	0.21	0.21	3.231	3.107	95.4	0.65
3.68	336.9	0.20	0.20	0.20	3.240	3.116	95.4	0.62
3.69	336.0	0.19	0.19	0.19	3.248	3.125	95.4	0.59
3.7	335.1	0.18	0.18	0.18	3.256	3.133	95.4	0.56
3.71	334.2	0.17	0.17	0.17	3.265	3.141	95.4	0.53
3.72	333.3	0.16	0.16	0.16	3.273	3.149	95.4	0.50
3.73	332.4	0.15	0.15	0.15	3.281	3.157	95.4	0.47
3.74	331.5	0.14	0.14	0.14	3.289	3.165	95.5	0.44
3.75	330.6	0.13	0.13	0.13	3.297	3.173	95.5	0.41
3.76	329.7	0.12	0.12	0.12	3.305	3.181	95.5	0.37
3.77	328.9	0.11	0.11	0.11	3.312	3.188	95.5	0.34
3.78	328.0	9.94E-02	9.94E-02	9.86E-02	3.320	3.196	95.5	0.32
3.79	327.1	9.08E-02	9.08E-02	9.01E-02	3.328	3.203	95.5	0.29
3.8	326.3	8.18E-02	8.18E-02	8.11E-02	3.335	3.211	95.5	0.26
3.81	325.4	7.36E-02	7.36E-02	7.30E-02	3.342	3.218	95.5	0.23
3.82	324.6	6.71E-02	6.71E-02	6.66E-02	3.349	3.225	95.5	0.21
3.83	323.7	6.11E-02	6.11E-02	6.06E-02	3.357	3.232	95.5	0.20
3.84	322.9	5.59E-02	5.59E-02	5.54E-02	3.364	3.240	95.5	0.18
3.85	322.0	5.11E-02	5.11E-02	5.07E-02	3.372	3.247	95.6	0.16

E_g [eV]	λ_g [nm]	J_{max} [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
3.86	321.2	4.60E-02	4.60E-02	4.57E-02	3.379	3.255	95.6	0.15
3.87	320.4	4.08E-02	4.08E-02	4.05E-02	3.386	3.261	95.6	0.13
3.88	319.5	3.63E-02	3.63E-02	3.60E-02	3.393	3.268	95.6	0.12
3.89	318.7	3.21E-02	3.21E-02	3.19E-02	3.400	3.275	95.6	0.10
3.9	317.9	2.82E-02	2.82E-02	2.80E-02	3.406	3.281	95.6	9.19E-02
3.91	317.1	2.45E-02	2.45E-02	2.43E-02	3.412	3.288	95.6	8.00E-02
3.92	316.3	2.13E-02	2.13E-02	2.11E-02	3.419	3.294	95.6	6.95E-02
3.93	315.5	1.87E-02	1.87E-02	1.85E-02	3.425	3.300	95.6	6.12E-02
3.94	314.7	1.60E-02	1.60E-02	1.59E-02	3.431	3.306	95.6	5.26E-02
3.95	313.9	1.35E-02	1.35E-02	1.34E-02	3.436	3.311	95.6	4.44E-02
3.96	313.1	1.13E-02	1.13E-02	1.12E-02	3.442	3.317	95.6	3.72E-02
3.97	312.3	9.28E-03	9.28E-03	9.21E-03	3.447	3.321	95.6	3.06E-02
3.98	311.5	7.47E-03	7.47E-03	7.41E-03	3.451	3.326	95.6	2.46E-02
3.99	310.7	5.89E-03	5.89E-03	5.85E-03	3.455	3.329	95.6	1.95E-02
4	310.0	4.67E-03	4.67E-03	4.64E-03	3.459	3.333	95.6	1.55E-02
4.01	309.2	3.80E-03	3.80E-03	3.78E-03	3.463	3.338	95.6	1.26E-02
4.02	308.4	3.02E-03	3.02E-03	2.99E-03	3.467	3.342	95.7	1.00E-02
4.03	307.7	2.29E-03	2.29E-03	2.27E-03	3.470	3.344	95.7	7.60E-03
4.04	306.9	1.69E-03	1.69E-03	1.68E-03	3.472	3.347	95.7	5.61E-03
4.05	306.1	1.27E-03	1.27E-03	1.26E-03	3.474	3.349	95.7	4.23E-03
4.06	305.4	9.26E-04	9.26E-04	9.19E-04	3.476	3.351	95.7	3.08E-03
4.07	304.6	6.32E-04	6.32E-04	6.28E-04	3.476	3.351	95.7	2.10E-03
4.08	303.9	4.29E-04	4.29E-04	4.26E-04	3.476	3.351	95.7	1.43E-03
4.09	303.1	2.72E-04	2.72E-04	2.70E-04	3.474	3.349	95.7	9.03E-04
4.1	302.4	1.66E-04	1.66E-04	1.65E-04	3.471	3.346	95.7	5.52E-04
4.11	301.7	1.08E-04	1.08E-04	1.07E-04	3.470	3.345	95.7	3.59E-04
4.12	300.9	6.61E-05	6.61E-05	6.56E-05	3.468	3.342	95.7	2.19E-04
4.13	300.2	4.14E-05	4.14E-05	4.10E-05	3.465	3.340	95.6	1.37E-04
4.14	299.5	2.40E-05	2.40E-05	2.39E-05	3.461	3.336	95.6	7.96E-05
4.15	298.8	1.37E-05	1.37E-05	1.36E-05	3.456	3.331	95.6	4.54E-05
4.16	298.0	7.52E-06	7.52E-06	7.46E-06	3.451	3.326	95.6	2.48E-05
4.17	297.3	3.76E-06	3.76E-06	3.73E-06	3.443	3.318	95.6	1.24E-05
4.18	296.6	1.88E-06	1.88E-06	1.86E-06	3.435	3.310	95.6	6.17E-06
4.19	295.9	8.53E-07	8.53E-07	8.46E-07	3.424	3.299	95.6	2.79E-06
4.2	295.2	3.44E-07	3.44E-07	3.41E-07	3.410	3.285	95.6	1.12E-06
4.21	294.5	1.36E-07	1.36E-07	1.35E-07	3.398	3.273	95.6	4.41E-07
4.22	293.8	5.68E-08	5.68E-08	5.64E-08	3.383	3.259	95.6	1.84E-07
4.23	293.1	1.92E-08	1.92E-08	1.91E-08	3.366	3.241	95.5	6.19E-08
4.24	292.4	6.95E-09	6.95E-09	6.89E-09	3.350	3.226	95.5	2.22E-08
4.25	291.7	2.33E-09	2.33E-09	2.31E-09	3.330	3.205	95.5	7.42E-09
4.26	291.0	5.44E-10	5.44E-10	5.39E-10	3.304	3.180	95.5	1.72E-09
4.27	290.4	1.53E-10	1.53E-10	1.51E-10	3.280	3.156	95.4	4.78E-10
4.28	289.7	3.89E-11	3.89E-11	3.85E-11	3.253	3.129	95.4	1.21E-10
4.29	289.0	6.40E-12	6.40E-12	6.35E-12	3.220	3.097	95.4	1.97E-11
4.3	288.3	1.44E-12	1.44E-12	1.43E-12	3.188	3.065	95.3	4.40E-12

E_g [eV]	λ_g [nm]	$J_{\max}$ [mA/cm ²]	J_{sc} [mA/cm ²]	J_{mpp} [mA/cm ²]	V_{oc} [V]	V_{mpp} [V]	FF [%]	η [%]
4.31	287.7	2.48E-13	2.48E-13	2.46E-13	3.149	3.026	95.3	7.48E-13
4.32	287.0	2.78E-14	2.78E-14	2.76E-14	3.110	2.988	95.2	8.23E-14
4.33	286.3	5.21E-15	5.21E-15	5.17E-15	3.072	2.950	95.2	1.53E-14
4.34	285.7	7.43E-16	7.43E-16	7.37E-16	3.032	2.910	95.1	2.15E-15
4.35	285.0	9.59E-17	9.59E-17	9.51E-17	2.993	2.871	95.1	2.73E-16
4.36	284.4	1.84E-17	1.84E-17	1.82E-17	2.957	2.836	95.0	5.18E-17
4.37	283.7	2.37E-18	2.37E-18	2.35E-18	2.904	2.783	95.0	6.60E-18
4.38	283.1	2.15E-19	2.15E-19	2.13E-19	2.865	2.745	94.9	5.85E-19
4.39	282.4	3.64E-20	3.64E-20	3.60E-20	2.829	2.709	94.9	9.77E-20
4.4	281.8	5.31E-21	5.31E-21	5.26E-21	2.780	2.661	94.8	1.41E-20
4.41	281.1	3.11E-22	3.11E-22	3.08E-22	2.703	2.584	94.7	8.09E-22
4.42	280.5	8.06E-24	8.06E-24	7.98E-24	2.644	2.526	94.6	2.02E-23